

Krishna Dheeraj Krovi

AI & Backend Software Engineer

krishnadheerajkrovi@gmail.com • Work Authorization: H1B • Open to Remote

Python · REST APIs & Microservices · LLM / Multi-Agent Systems · RAG · AWS & Azure · Docker / K8s · 4 Years Experience

SUMMARY

Delivery-focused Software Engineer with 4 years of experience leading the development of scalable backend systems and AI-powered platforms. Proven expertise in writing clean, maintainable code across multiple languages (Python, Go, Java) and integrating generative AI tools into enterprise solutions. Drives the full software development lifecycle from solution design and automated testing to CI/CD and observability, ensuring alignment with platform and business objectives.

SKILLS

Languages: Python, Go, SQL, Java, JavaScript, Bash

Backend: RESTful API Design, Microservices, MVC Architecture, Design Patterns, Server-Side Development, System Integration

Databases: PostgreSQL, MySQL, SQL Server, Oracle; Query Optimization, Indexing, Data Modeling

Async / Caching: Redis, Task Queues, Asynchronous Processing, Event-Driven Processing, Spark

Cloud & DevOps: AWS, Kubernetes, Docker, CI/CD (GitHub Actions, ArgoCD), Observability (MLflow, OpenTelemetry, CloudWatch)

Quality & Testing: Automated Testing (pytest, TestNG, Mockito), Test-Driven Development (TDD), Code Review, Debugging, Defect Management

AI / LLM & Data: Gen AI Tools (OpenAI, Anthropic, Gemini), Multi-Agent Systems, RAG, Vector DBs (Milvus, pgvector), LangChain, LangGraph, PySpark

WORK EXPERIENCE

Artha Solutions · Software Engineer

May 2022 - Present

- Architected a multi agent LLM conversion pipeline with a **deterministic, code-based verifier** using AST, lint and rule gate, making agent output reproducible, gate-enforced and auditable
- Built the execution on **AWS Step Functions** and **Bedrock AgentCore** with checkpointed resumable steps, human-in-the-loop escalation, bounded retry with stall detection, and a per-job token ceiling, eliminating runaway loops and unbounded LLM spend
- Operationalized **Claude Code** as a governed engineering workflow with enforced agent rules, repo-scoped skills, knowledge-graph retrieval hooks gated by CI with pytest, ruff, bandit, gitleaks, pip-audit and SHA-pinned actions, ensuring no AI-generated code deploys ungated
- Delivered **50+ RESTful APIs** for an enterprise platform migration at **100%** functional integrity, backed by **PostgreSQL**, with robust error handling, request validation, and structured logging
- Improved query performance through **indexing**, selective denormalization, and schema redesign across PostgreSQL and MySQL, measurably reducing job runtimes on high-volume workloads
- Applied **asynchronous processing** and **Redis** caching with Spark parallelization on an Elasticsearch service indexing **1.5M records**, cutting backend response times by **50%**
- Architected **microservice** boundaries and shared libraries to decompose a monolithic workflow into independently deployable services, improving maintainability and release cadence
- Deployed and operated services on **AWS (EC2, S3, RDS, Lambda)** with Infrastructure-as-Code (CloudFormation) and Secrets Manager, containerized via **Docker and Kubernetes**
- Owned **CI/CD** pipelines across **Jenkins, GitHub Actions, and Bitbucket Pipelines** (OIDC auth, RBAC), improving release reliability and reducing manual deployment effort **4x**
- Promoted engineering excellence and mentored peers by implementing deterministic quality ship gates with **16** automated checks, allowing additional safety in deployment of LLM-generated code
- Designed backend integrations and data-synchronization services (SAP ERP, CMMS) with scheduled jobs and change-data handling for reliable cross-system consistency
- Hardened **API security** with rate limiting, timeout handling, request validation, and structured logging; built background workers and task queues with checkpoint/resume to offload long-running jobs

Wipro Limited · Project Engineer

Dec 2020 - Jun 2021

- Developed and tested **REST API** integrations (Java, Postman) on an **MVC** architecture, improving service validation and reducing manual effort
- Wrote **SQL** queries and data-extraction workflows for reporting, operational analysis, and downstream application processes
- Supported production issue resolution, release deployments, and Agile delivery, including defect tracking, UAT support, and sprint-based execution

EDUCATION

M.S., Electrical Engineering, University of Southern California, Los Angeles, USA

M.Tech & B.Tech, Electronics & Communications Engineering, JNTU, Hyderabad, India

CERTIFICATIONS

- AWS Certified Machine Learning - Specialty, Amazon Web Services
- Claude Certified Architect - Foundations, Anthropic